

PM Routing Interpretability

which pathways are kept vs. dropped

Cohort: `pan_meta_pri` (`mut_cnv`) — 8893 samples, 1258 Reactome pathway tokens, 2 classes.

Protocol: unified 5-fold CV, seed 42 (20% test / 10%-of-train val), identical folds to Table 2. Model: canonical bioMoR (`bio_both`, learned pathway graph at both sites, sum-pooled).

Keep/drop signal: each pathway token’s *mean recursion depth* over held-out test folds. Under **expert-choice** a capacity funnel keeps a shrinking top- k each step ($\text{depth} \in [0, K] = \text{steps survived}$); under **token-choice** each pathway self-gates one $\text{depth} \in [1, K]$. Higher depth = router **keeps** compute; the minimum = **dropped** (exits early).

Summary

Routing	K	Macro-F1	mean depth	active tokens / step	kept-to- K
expert	1	85.1±1.3	1.00	1258	1258/1258
expert	2	84.7±2.8	1.75	1258, 944	1189/1258
expert	3	85.9±2.2	2.50	1258, 944, 944	684/1258
expert	4	83.9±1.9	3.25	1258, 944, 944, 944	379/1258
token	1	85.5±2.5	1.00	1258	1258/1258
token	2	86.1±1.5	1.49	1258, 611	604/1258
token	3	84.5±4.2	2.02	1258, 846, 436	115/1258
token	4	85.3±3.6	2.51	1258, 942, 629, 323	36/1258

expert-choice, $K = 1$ (Macro-F1 85.1±1.3)

$K=1$ is a single pass — every pathway is kept to depth 1, so there is no keep/drop decision.

expert-choice, $K = 2$ (Macro-F1 84.7±2.8)

KEPT — top 25 deepest-routed

#	Pathway	depth
1	R-HSA-2559583	1.99
2	R-HSA-2990846	1.99
3	R-HSA-69563	1.99
4	R-HSA-5358346	1.99
5	R-HSA-163685	1.99
6	R-HSA-157118	1.98
7	R-HSA-5083625	1.98
8	R-HSA-9755511	1.98
9	R-HSA-2559580	1.98
10	R-HSA-1236974	1.98
11	R-HSA-9711123	1.98
12	R-HSA-388841	1.98
13	R-HSA-163560	1.98
14	R-HSA-212165	1.98
15	R-HSA-936964	1.98
16	R-HSA-3000171	1.97
17	R-HSA-5632684	1.97
18	R-HSA-9917777	1.97
19	R-HSA-5579029	1.97
20	R-HSA-9759194	1.97
21	R-HSA-1980145	1.97
22	R-HSA-1989781	1.97
23	R-HSA-9613829	1.97
24	R-HSA-8982491	1.97
25	R-HSA-9843745	1.97

DROPPED — 25 earliest-exit

#	Pathway	depth
1	R-HSA-9629569	1.09
2	R-HSA-450513	1.20
3	R-HSA-180024	1.22
4	R-HSA-450385	1.27
5	R-HSA-450604	1.31
6	R-HSA-3560783	1.34
7	R-HSA-210500	1.34
8	R-HSA-392517	1.34
9	R-HSA-110330	1.35
10	R-HSA-450302	1.36
11	R-HSA-5651801	1.37
12	R-HSA-9920588	1.37
13	R-HSA-9675135	1.38
14	R-HSA-176187	1.38
15	R-HSA-111471	1.38
16	R-HSA-76071	1.38
17	R-HSA-1810476	1.39
18	R-HSA-5625740	1.39
19	R-HSA-9709570	1.40
20	R-HSA-844456	1.40
21	R-HSA-5601884	1.41
22	R-HSA-3560782	1.41
23	R-HSA-9735871	1.41
24	R-HSA-1606322	1.41
25	R-HSA-114452	1.42

expert-choice, $K = 3$ (Macro-F1 85.9 ± 2.2)

KEPT — top 25 deepest-routed

#	Pathway	depth
1	R-HSA-3229121	2.99
2	R-HSA-205043	2.99
3	R-HSA-6803204	2.99
4	R-HSA-6782315	2.98
5	R-HSA-9764725	2.98
6	R-HSA-6793080	2.98
7	R-HSA-9920951	2.97
8	R-HSA-9772755	2.97
9	R-HSA-5654219	2.97
10	R-HSA-9615710	2.97
11	R-HSA-192823	2.97
12	R-HSA-211976	2.97
13	R-HSA-9619665	2.96
14	R-HSA-9764560	2.96
15	R-HSA-204998	2.96
16	R-HSA-917729	2.96
17	R-HSA-2173796	2.96
18	R-HSA-196757	2.96
19	R-HSA-1502540	2.95
20	R-HSA-5083625	2.95
21	R-HSA-8982491	2.95
22	R-HSA-4641263	2.95
23	R-HSA-6804757	2.95
24	R-HSA-5633007	2.95
25	R-HSA-9758274	2.94

DROPPED — 25 earliest-exit

#	Pathway	depth
1	R-HSA-3928664	1.52
2	R-HSA-8862803	1.67
3	R-HSA-3296482	1.68
4	R-HSA-173623	1.69
5	R-HSA-2514856	1.70
6	R-HSA-5668599	1.73
7	R-HSA-452723	1.74
8	R-HSA-203927	1.75
9	R-HSA-5627123	1.77
10	R-HSA-4419969	1.77
11	R-HSA-5601884	1.78
12	R-HSA-1482788	1.78
13	R-HSA-6803529	1.79
14	R-HSA-1482839	1.80
15	R-HSA-211000	1.82
16	R-HSA-1482801	1.84
17	R-HSA-9768727	1.84
18	R-HSA-352230	1.87
19	R-HSA-1482925	1.87
20	R-HSA-5669034	1.87
21	R-HSA-1482922	1.87
22	R-HSA-73933	1.88
23	R-HSA-606279	1.89
24	R-HSA-450302	1.90
25	R-HSA-9659787	1.90

expert-choice, $K = 4$ (Macro-F1 83.9±1.9)

KEPT — top 25 deepest-routed

#	Pathway	depth
1	R-HSA-1632852	4.00
2	R-HSA-2028269	3.99
3	R-HSA-74751	3.99
4	R-HSA-9663891	3.99
5	R-HSA-2565942	3.99
6	R-HSA-9678108	3.99
7	R-HSA-9706574	3.99
8	R-HSA-2404192	3.99
9	R-HSA-442982	3.99
10	R-HSA-8963899	3.99
11	R-HSA-918233	3.99
12	R-HSA-5628897	3.99
13	R-HSA-3769402	3.98
14	R-HSA-9013418	3.98
15	R-HSA-9833110	3.98
16	R-HSA-445989	3.98
17	R-HSA-499943	3.98
18	R-HSA-2559580	3.98
19	R-HSA-168898	3.98
20	R-HSA-109704	3.98
21	R-HSA-9818027	3.98
22	R-HSA-442742	3.98
23	R-HSA-5693571	3.98
24	R-HSA-1566977	3.98
25	R-HSA-69206	3.98

DROPPED — 25 earliest-exit

#	Pathway	depth
1	R-HSA-977606	1.73
2	R-HSA-166663	1.74
3	R-HSA-166658	1.81
4	R-HSA-202433	1.81
5	R-HSA-2029481	1.84
6	R-HSA-391160	1.94
7	R-HSA-6803529	1.97
8	R-HSA-1483206	1.99
9	R-HSA-3371568	2.01
10	R-HSA-9664323	2.05
11	R-HSA-111465	2.19
12	R-HSA-9828806	2.20
13	R-HSA-9830364	2.21
14	R-HSA-4419969	2.21
15	R-HSA-202131	2.23
16	R-HSA-1483166	2.24
17	R-HSA-9820965	2.24
18	R-HSA-2173788	2.24
19	R-HSA-171319	2.24
20	R-HSA-606279	2.25
21	R-HSA-1482925	2.25
22	R-HSA-8862803	2.25
23	R-HSA-76071	2.25
24	R-HSA-212300	2.25
25	R-HSA-352230	2.26

token-choice, $K = 1$ (Macro-F1 85.5 ± 2.5)

$K=1$ is a single pass — every pathway is kept to depth 1, so there is no keep/drop decision.

token-choice, $K = 2$ (Macro-F1 86.1 ± 1.5)

KEPT — top 25 deepest-routed

#	Pathway	depth
1	R-HSA-9920951	2.00
2	R-HSA-9931510	1.99
3	R-HSA-9821993	1.99
4	R-HSA-379716	1.99
5	R-HSA-75876	1.99
6	R-HSA-140534	1.99
7	R-HSA-9940465	1.99
8	R-HSA-9648002	1.99
9	R-HSA-9970672	1.99
10	R-HSA-5357769	1.99
11	R-HSA-912446	1.99
12	R-HSA-5654228	1.98
13	R-HSA-5654719	1.98
14	R-HSA-5654712	1.98
15	R-HSA-937072	1.98
16	R-HSA-71336	1.98
17	R-HSA-5690714	1.98
18	R-HSA-428930	1.98
19	R-HSA-982772	1.98
20	R-HSA-9733458	1.98
21	R-HSA-75105	1.98
22	R-HSA-8849932	1.98
23	R-HSA-9705462	1.97
24	R-HSA-8875360	1.97
25	R-HSA-392170	1.97

DROPPED — 25 earliest-exit

#	Pathway	depth
1	R-HSA-453274	1.00
2	R-HSA-3858494	1.00
3	R-HSA-170834	1.01
4	R-HSA-9006936	1.01
5	R-HSA-9010553	1.01
6	R-HSA-195721	1.01
7	R-HSA-4641258	1.01
8	R-HSA-201681	1.01
9	R-HSA-6791312	1.01
10	R-HSA-2467813	1.01
11	R-HSA-376176	1.02
12	R-HSA-349425	1.02
13	R-HSA-373755	1.02
14	R-HSA-174178	1.02
15	R-HSA-432722	1.02
16	R-HSA-8873719	1.02
17	R-HSA-9948299	1.02
18	R-HSA-174143	1.02
19	R-HSA-6811434	1.02
20	R-HSA-72312	1.03
21	R-HSA-450408	1.03
22	R-HSA-9648025	1.03
23	R-HSA-3371556	1.03
24	R-HSA-162909	1.03
25	R-HSA-3928665	1.03

token-choice, $K = 3$ (Macro-F1 84.5 ± 4.2)

KEPT — top 25 deepest-routed

#	Pathway	depth
1	R-HSA-9648002	2.99
2	R-HSA-5358508	2.84
3	R-HSA-9687136	2.81
4	R-HSA-9709570	2.78
5	R-HSA-9755779	2.78
6	R-HSA-5689901	2.77
7	R-HSA-6783310	2.77
8	R-HSA-174048	2.76
9	R-HSA-936964	2.75
10	R-HSA-2243919	2.74
11	R-HSA-6802946	2.74
12	R-HSA-450321	2.74
13	R-HSA-3000157	2.73
14	R-HSA-9675126	2.72
15	R-HSA-977225	2.71
16	R-HSA-2214320	2.71
17	R-HSA-442660	2.71
18	R-HSA-176412	2.71
19	R-HSA-3134975	2.70
20	R-HSA-174437	2.70
21	R-HSA-179409	2.70
22	R-HSA-6805567	2.69
23	R-HSA-6809371	2.69
24	R-HSA-166663	2.68
25	R-HSA-9823730	2.68

DROPPED — 25 earliest-exit

#	Pathway	depth
1	R-HSA-170834	1.01
2	R-HSA-8853659	1.09
3	R-HSA-5625900	1.12
4	R-HSA-180292	1.14
5	R-HSA-383280	1.14
6	R-HSA-9006936	1.21
7	R-HSA-5627123	1.22
8	R-HSA-977444	1.23
9	R-HSA-9665348	1.24
10	R-HSA-5601884	1.24
11	R-HSA-418038	1.25
12	R-HSA-5617472	1.25
13	R-HSA-2173789	1.25
14	R-HSA-375280	1.26
15	R-HSA-5610787	1.26
16	R-HSA-8941326	1.26
17	R-HSA-977443	1.27
18	R-HSA-201681	1.28
19	R-HSA-5578775	1.28
20	R-HSA-9013404	1.29
21	R-HSA-674695	1.29
22	R-HSA-9006115	1.30
23	R-HSA-167242	1.31
24	R-HSA-418597	1.31
25	R-HSA-451326	1.31

token-choice, $K = 4$ (Macro-F1 85.3 ± 3.6)

KEPT — top 25 deepest-routed

#	Pathway	depth
1	R-HSA-982772	3.82
2	R-HSA-9828806	3.82
3	R-HSA-9836573	3.80
4	R-HSA-977068	3.76
5	R-HSA-9820965	3.75
6	R-HSA-9970672	3.74
7	R-HSA-9821002	3.71
8	R-HSA-9931295	3.71
9	R-HSA-5334118	3.70
10	R-HSA-937072	3.66
11	R-HSA-8931838	3.66
12	R-HSA-9710421	3.65
13	R-HSA-196807	3.65
14	R-HSA-199220	3.64
15	R-HSA-3238698	3.64
16	R-HSA-9821993	3.61
17	R-HSA-187687	3.61
18	R-HSA-1810476	3.61
19	R-HSA-9619483	3.60
20	R-HSA-9755779	3.58
21	R-HSA-5635838	3.58
22	R-HSA-1912420	3.58
23	R-HSA-445144	3.58
24	R-HSA-9037629	3.57
25	R-HSA-70221	3.57

DROPPED — 25 earliest-exit

#	Pathway	depth
1	R-HSA-170834	1.28
2	R-HSA-68949	1.36
3	R-HSA-2565942	1.41
4	R-HSA-373076	1.42
5	R-HSA-9006936	1.42
6	R-HSA-5673001	1.42
7	R-HSA-5655302	1.44
8	R-HSA-909733	1.44
9	R-HSA-9637690	1.44
10	R-HSA-69052	1.45
11	R-HSA-1643713	1.46
12	R-HSA-8878159	1.47
13	R-HSA-2173793	1.49
14	R-HSA-5633008	1.49
15	R-HSA-5668541	1.50
16	R-HSA-2559586	1.50
17	R-HSA-9948299	1.52
18	R-HSA-418038	1.54
19	R-HSA-380320	1.55
20	R-HSA-9820952	1.55
21	R-HSA-9680350	1.56
22	R-HSA-69306	1.57
23	R-HSA-8864260	1.58
24	R-HSA-1483257	1.58
25	R-HSA-5683057	1.58